LABORATORY RECORD

Course	Full Stack Web Development
Course Code	23CM4121
Experiment No.	4
Student Name	M. Saicharan
Roll No.	A24126552096
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1. TITLE

Design and Implementation of an Interactive Web Portal Using HTML, CSS and JavaScript DOM Manipulation

2. AIM

To design and develop a dynamic interactive webpage using HTML, CSS, and JavaScript with user greeting, live digital time, dynamic shopping-list creation, and background theme switching.

3. OBJECTIVE

1. To create the basic webpage structure using HTML.
2. To style the webpage using CSS gradients, cards, buttons, shadows, and spacing.
3. To access HTML elements using document.getElementById().
4. To display a personalized greeting based on user input.
5. To implement a live digital clock using Date() and setInterval().
6. To dynamically create and append list items using JavaScript.
7. To modify webpage styles dynamically using JavaScript functions.

4. THEORY

JavaScript makes webpages interactive by manipulating the Document Object Model (DOM). The DOM represents HTML elements as objects that JavaScript can access and modify. The getElementById() method selects an element by its ID, while innerHTML changes its displayed content. createElement() creates a new HTML element and appendChild() inserts it into the document. The Date object provides the current time, and setInterval() repeatedly executes a function after a specified interval. JavaScript can also modify CSS properties through the style object, allowing the webpage theme to change dynamically.

Application Flow: Load Webpage → Accept Name → Display Greeting → Update Live Time → Add Shopping Items → Change Background Theme

5. ALGORITHM / PROCEDURE

1. Create an HTML document with the title 'Interactive Web Portal'.
2. Define CSS styles for the page, header, container, panels, inputs, buttons, and list items.
3. Create a greeting panel with a name input and Greet button.
4. Create a digital time panel containing an element to display the current time.
5. Create a shopping list panel with an input field, Add Item button, and unordered list.
6. Create a background theme panel with Purple, Orange, and Night buttons.
7. Define showGreeting() to read the entered name and display a welcome message.
8. Define updateTime() using Date().toLocaleTimeString() and call it every second using setInterval().

9. Define addItem() to create a new list item and append it to the shopping list.
10. Define theme functions to dynamically change the body's background gradient.
11. Run the webpage in a browser and verify all interactive functions.

6. PROGRAM

dynamic.html

```
<!DOCTYPE html>
<html>
<head>

<title>Interactive Web Portal</title>

<style>

* {
margin: 0;
padding: 0;
box-sizing: border-box;
}

body {
font-family: Arial;
background: linear-gradient(135deg, #667eea, #764ba2);
min-height: 100vh;
}

header {
background: #111;
color: white;
text-align: center;
padding: 20px;
}

.container {
width: 95%;
max-width: 1200px;
margin: 20px auto;
}

.panel {
background: white;
padding: 30px;
min-height: 180px;
margin-bottom: 20px;
border-radius: 12px;
box-shadow: 0 5px 15px #555;
}

input {
padding: 12px;
width: 65%;
margin: 15px 0;
}

button {
padding: 10px 15px;
border: none;
border-radius: 5px;
background: #e83e8c;
color: white;
cursor: pointer;
}

button:hover {
background: #c2185b;
}

#time {
font-size: 35px;
margin-top: 25px;
}

li {
margin: 8px;
padding: 8px;
background: #eee;
}

</style>
</head>

<body>

<header>
<h1>Interactive Web Portal</h1>
<p>HTML, CSS and JavaScript Project</p>
```

```
</header>

<div class="container">

<div class="panel">
<h2>Greeting Section</h2>
<input id="userName" placeholder="Enter your name">
<button onclick="showGreeting()">Greet</button>
<p id="greeting"></p>
</div>

<div class="panel">
<h2>Digital Time</h2>
<p id="time"></p>
</div>

<div class="panel">
<h2>Shopping List</h2>
<input id="item" placeholder="Enter item">
<button onclick="addItem()">Add Item</button>
<ul id="items"></ul>
</div>

<div class="panel">
<h2>Background Theme</h2>
<button onclick="purpleTheme()">Purple</button>
<button onclick="orangeTheme()">Orange</button>
<button onclick="nightTheme()">Night</button>
</div>

</div>

<script>

function showGreeting() {
let name = document.getElementById("userName").value;
document.getElementById("greeting").innerHTML =
"Hello " + name + "! Welcome to the portal.";
}

function updateTime() {
document.getElementById("time").innerHTML =
new Date().toLocaleTimeString();
}

setInterval(updateTime, 1000);
updateTime();

function addItem() {
let itemName = document.getElementById("item").value;

if (itemName != "") {
let li = document.createElement("li");
li.innerHTML = itemName;
document.getElementById("items").appendChild(li);
}
}

function purpleTheme() {
document.body.style.background =
"linear-gradient(135deg,#667eea,#764ba2)";
}

function orangeTheme() {
document.body.style.background =
"linear-gradient(135deg,#ff9966,#ff5e62)";
}

function nightTheme() {
document.body.style.background =
"linear-gradient(135deg,#141e30,#243b55)";
}

</script>

</body>
</html>
```

7. CODE EXPLANATION

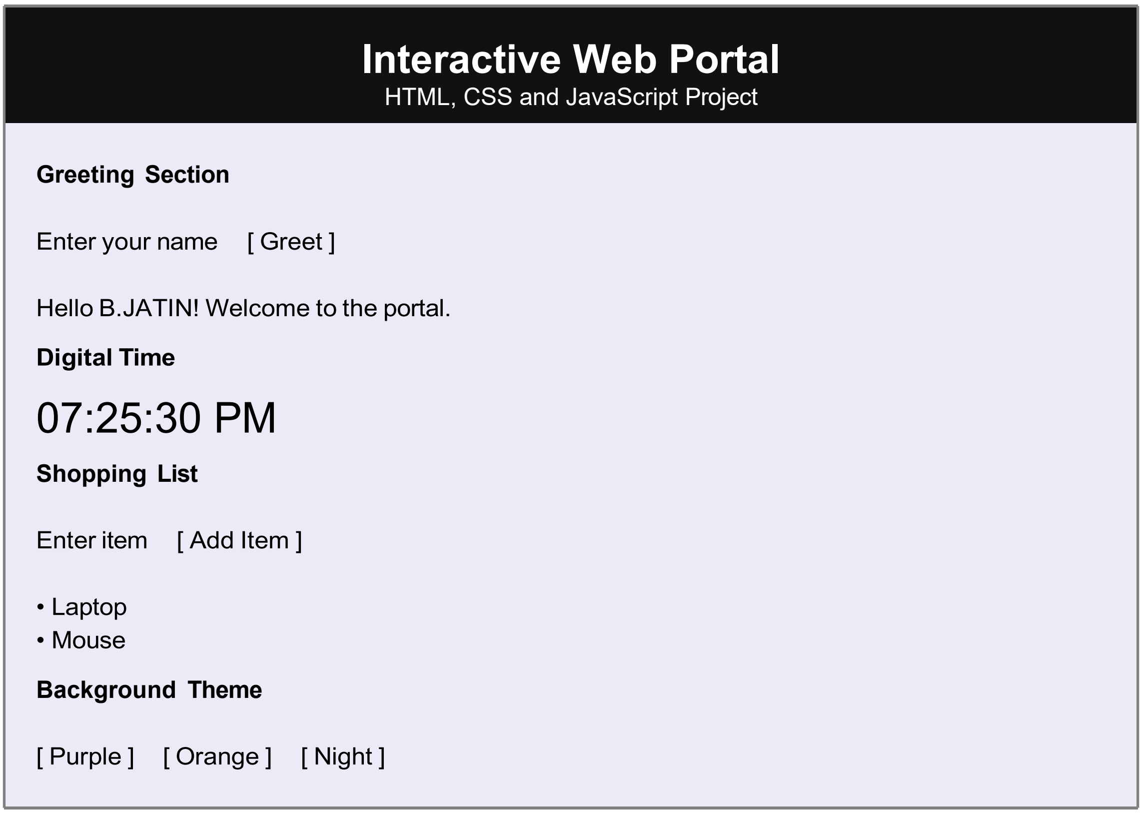
Code / Syntax	Explanation
document.getElementById()	Selects an HTML element using its unique ID.
innerHTML	Reads or changes the content displayed inside an HTML element.
new Date()	Creates a Date object containing the current date and time.
setInterval()	Executes updateTime() repeatedly every 1000 milliseconds.
document.createElement('li')	Creates a new list item dynamically.
appendChild()	Adds the newly created list item to the shopping list.
document.body.style.background	Changes the page background dynamically through JavaScript.
onclick	Calls a JavaScript function when the user clicks a button.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Enter B.JATIN and click Greet	Greeting message is displayed	Displayed correctly	Pass
TC-02	Observe Digital Time	Time updates every second	Updates correctly	Pass
TC-03	Enter Laptop and click Add Item	Laptop is added to the list	Added correctly	Pass
TC-04	Click Purple	Purple gradient is applied	Theme changed	Pass
TC-05	Click Orange	Orange gradient is applied	Theme changed	Pass
TC-06	Click Night	Dark gradient is applied	Theme changed	Pass

9. OUTPUT

Output 1: Interactive Web Portal with greeting, digital clock, shopping list, and theme controls.



10. RESULT

Thus, the Interactive Web Portal was successfully designed and implemented using HTML, CSS, and JavaScript. The program successfully demonstrated DOM manipulation through personalized greeting messages, a live digital clock, dynamic shopping-list creation, and background theme changes.